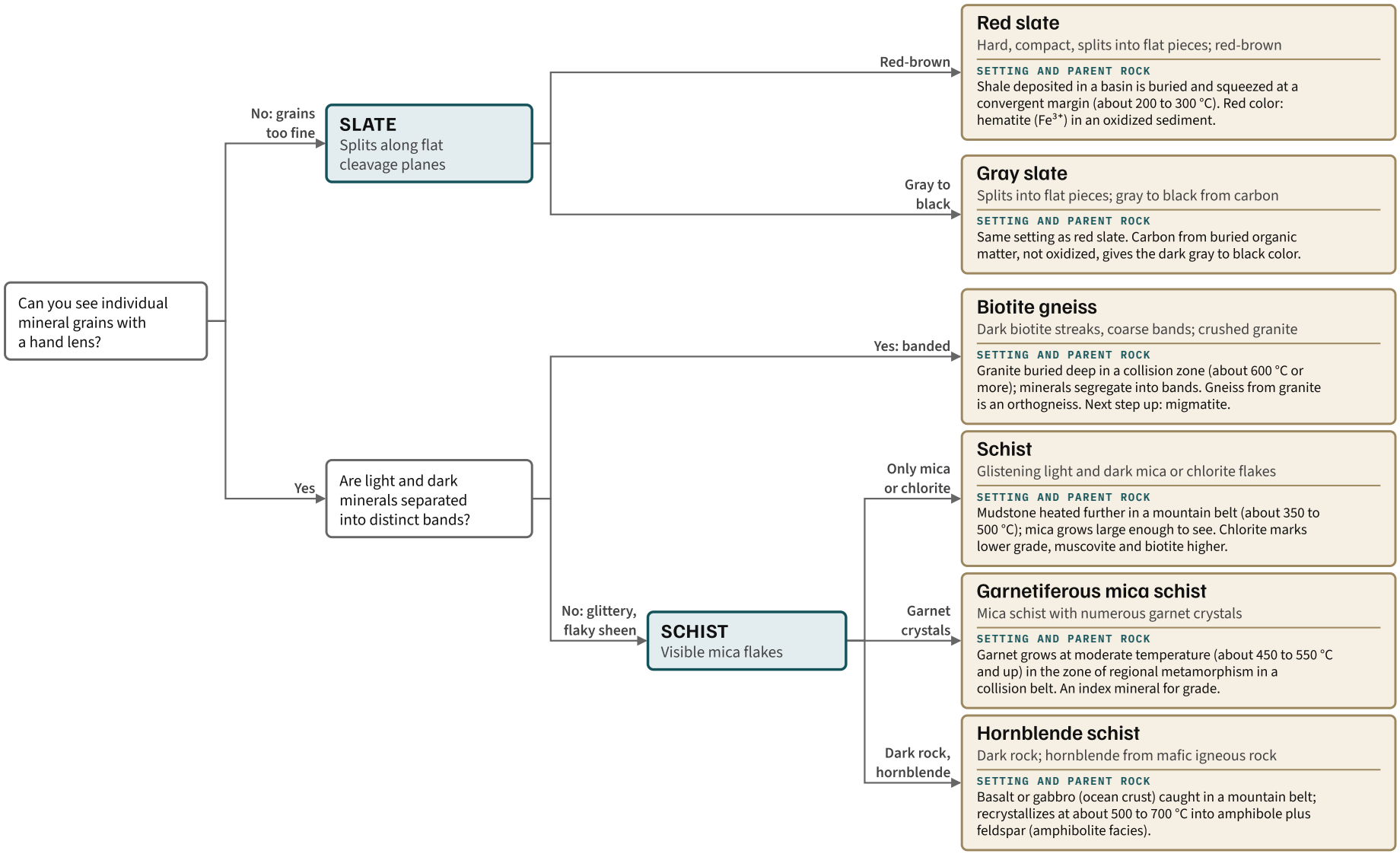
Foliated rocks

Use this chart when the minerals are aligned, flaky or in bands, so the rock looks streaked or splits in one direction.



Foliated rocks and metamorphic grade

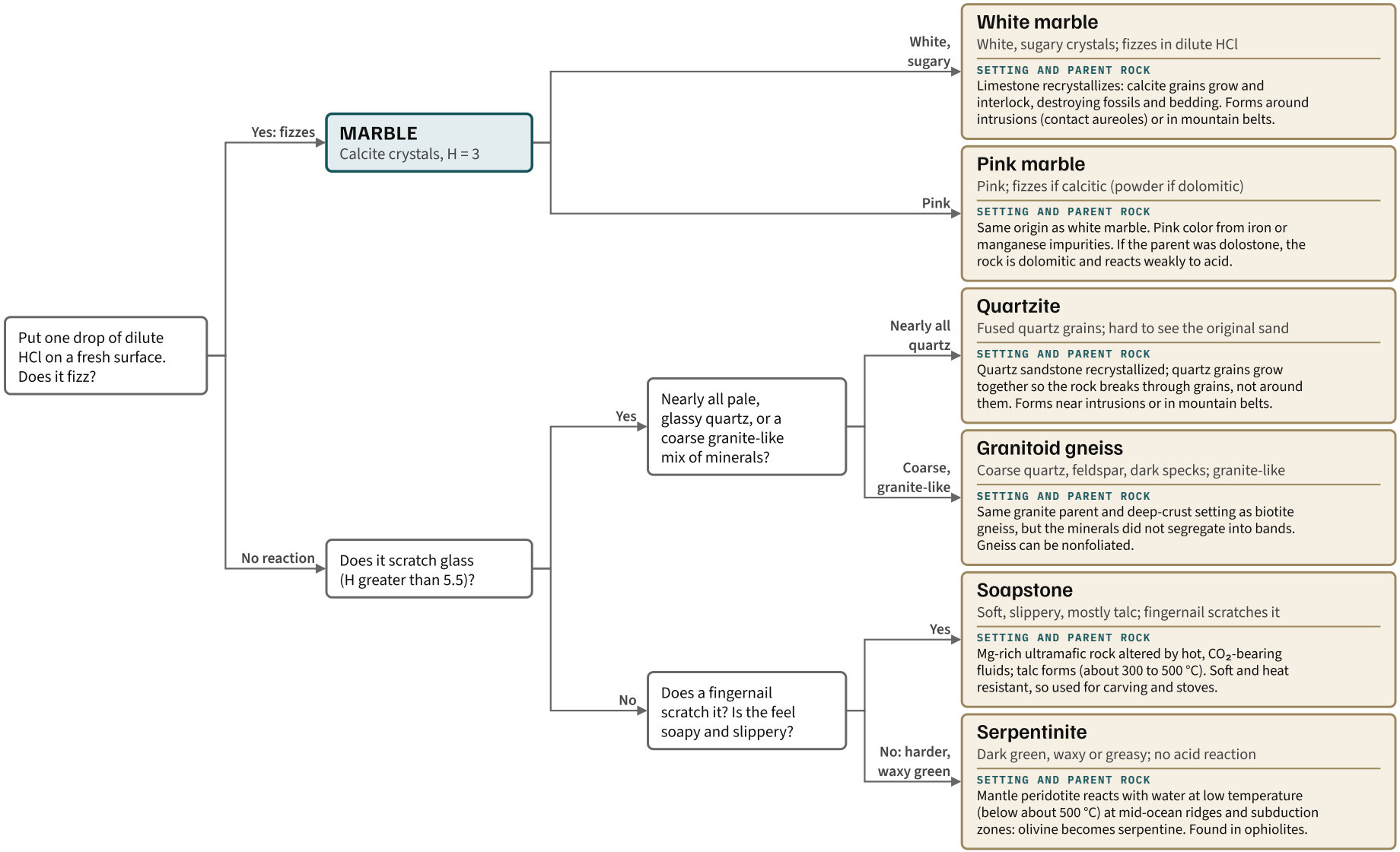
Slate, phyllite, schist and gneiss form a sequence of increasing temperature and pressure. Grains coarsen and minerals separate into bands. Phyllite is not in this set: grains are still too fine to see, but it has a satiny sheen and wavy surfaces.

Parent rocks

Slate and schist form from shale, biotite gneiss from granite, and hornblende schist from basalt or gabbro. Granitoid gneiss, which is nonfoliated in this collection, is on Chart 2.

Nonfoliated rocks

Use this chart when the rock is massive, with interlocking crystals and no layering or preferred direction.



Hardness benchmarks (Mohs) and acid test

Fingernail 2.5, copper coin about 3.5, steel nail and glass about 5.5, quartz 7. One drop of dilute HCl, goggles on: calcite marble fizzes; dolomitic marble needs scratched powder first.

Parent rocks

Marble forms from limestone, quartzite from sandstone, granitoid gneiss from granite, and serpentinite and soapstone from ultramafic rock such as peridotite.